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ABSTRACT

The composition and evaluation of a mushroom patty are composed of the following ingredients: oyster mushroom, the main ingredient that is locally available and rich in nutritional factors. All-purpose flour and egg are added to serve as a stabilizer and binder of the product. Onion leeks and Carrots were added to serve as an extender—salt to enhance its flavor, and pepper as an additional flavor. The experimental development research generally aimed to prepare a mushroom patty to replace the existing meat patty products and determine its acceptability. Specifically, it sought to (1) Determine the level of acceptability of mushroom patty and determine the sensory properties of the oyster mushroom patty in terms of taste, texture, aroma, color, and general acceptability; (2) Determine the consumers' acceptability of oyster mushroom patty; and (3) Determine the cost of production of the oyster mushroom patty. Analysis of variance in Completely Randomized Design (CRD) was used to analyze the data, and a significant difference occurred. In the three oyster mushroom formulations, it was found that 400g of oyster mushroom was the most acceptable product as perceived by the consumers. The product was described as "moderate brown" in terms of color, "like slightly mushroom fragrance" in terms of aroma, "like moderate mushroom flavor" when it comes to taste, and "like moderate appearance" in terms of texture. The Hedonic test results highlighted that 300g of oyster mushroom had the best color; 300g, 350g, and 400g of oyster mushroom had the same result of like slightly mushroom in terms of aroma; 350g and 400g of oyster mushroom were the most palatable and exhibit better texture; and 400g of oyster mushroom had the highest general acceptability. The sensory result is subjected to consumer acceptability. The employee respondent was rated "like very much." There are significant differences existing among the three treatments of mushroom patty.

Keywords;

Oyster Mushroom, Mushroom Patty, Patty, Plant Based food

INTRODUCTION

The oyster mushroom, scientifically known as *Pleurotus ostreatus*, is a species of edible mushroom widely cultivated and enjoyed for its culinary value. Oyster mushrooms have a distinct appearance with fan-shaped or oyster-shell-shaped caps. The cap can vary in color from white to gray to brown, depending on the variety. The gills on the underside of the cap are white and run down the short stem.

In developing third-world nations, malnutrition is a problem. Mushrooms have been identified as great food sources to reduce malnutrition in underdeveloped countries because of their flavor, texture, nutritional content, and high productivity per unit area.

In Asia and many other regions, mushrooms have been used for their therapeutic and gastronomic benefits. Approximately 5,000 different species of

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mushrooms, of which at least 1220 are reported to be edible (*Muhammad Nasir et al., 2006*). Given that edible mushroom extracts have a lot of bioactive chemicals, there is much interest in using them as dietary supplements. Mushrooms' early adoption is due in part to their high nutrient content. For flavor and as a meat alternative, people eat mushrooms. Mushrooms that are low in fat and calories, high in vitamins B and C, have the most protein of any other food that comes from plants, and are an excellent source of minerals.

This study intends to provide a patty utilized by oyster mushrooms to create an additional product, extend the shelf life, and increase the production of mushrooms in the province of Capiz.

OBJECTIVES

This study aims to;

1. Determine the level of acceptability of mushroom patty and determine the sensory properties of the oyster mushroom patty in terms of taste, texture, aroma, color, and general acceptability;
2. Determine the consumers' acceptability of oyster mushroom patty; and
3. Determine the cost of production of the oyster mushroom patty.

METHODOLOGY

This study was conducted using developmental research at the Food and Fisheries Research Development Center at Capiz State University – Dayao Satellite College, Dayao, Roxas, and Capiz. The experiment was used to investigate the acceptability of oyster mushrooms as a patty.

Completely Randomized Design (CRD) was utilized in this study wherein oyster mushrooms were used in making patty and subsequently replicated.

The proportion of ingredients per treatment. Treatment A with 300g of oyster mushroom; Treatment B with 350g of oyster mushroom; and Treatment C with 400g Oyster Mushroom. Other ingredients are constant.

Ingredients	Treatment A	Treatment B	Treatment C
<i>Oyster Mushroom</i>	300g	350g	400g
<i>All-purpose Flour</i>	150g	150g	150g
<i>Carrot</i>	20g	20g	20g
<i>Onion leeks</i>	10g	10g	10g
<i>Garlic</i>	5g	5g	5g
<i>Onion</i>	10g	10g	10g
<i>Salt</i>	7g	7g	7g
<i>White Pepper</i>	1.5g	1.5g	1.5g
<i>Egg</i>	Medium	Medium	Medium
<i>Sugar</i>	5g	5g	5g

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Scoring the sensory quality and acceptability of oyster mushroom patty regarding with color, taste, flavor, texture, and general acceptability— a nine-point hedonic scale for the sensory characteristics and evaluated by 30 evaluators. A nine-point hedonic scale will be used for consumers' preferences, and 100 respondents will determine the products.

The data was gathered after every replication and was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

Scoring of Variables

In the scoring variables of this study, a nine–point hedonic scoring scale was used in evaluating the product. In this study, the sensory characteristics such as color, aroma, taste, texture, and general acceptability of oyster mushroom patty were scored, given weight, and categorized.

Table 1.0. *Nine-Point Hedonic Soring Scale in terms of Color.*

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
9	Very Dark Brown 	8.12 – 9.0
8	Dark Brown 	7.23 – 8.11
7	Medium Brown 	6.34 – 7.22
6	Light Brown 	5.45 – 6.33
5	Dark Blond 	4.56 – 5.44
4	Medium Blond 	3.67 – 4.55
3	Light Blond 	2.78 – 3.66
2	Very Light Blond 	1.89 – 2.77
1	Lightest Blond 	1.00 – 1.88

Table 1.1. *Nine-Point Hedonic Soring Scale in terms of Aroma.*

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
9	Like Extremely Mushroom aroma	8.12 – 9.0
8	Like very much Mushroom aroma	7.23 – 8.11
7	Like moderately Mushroom aroma	6.34 – 7.22
6	Like slightly Mushroom aroma	5.45 – 6.33
5	Neither like nor dislike Mushroom aroma	4.56 – 5.44
4	Dislike slightly Mushroom aroma	3.67 – 4.55
3	Dislike moderately Mushroom aroma	2.78 – 3.66
2	Dislike very much Mushroom aroma	1.89 – 2.77
1	Dislike extremely Mushroom aroma	1.00 – 1.88

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Table 1.2. *Nine -Point Hedonic Soring Scale in terms of Taste.*

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
9	Like Extremely Mushroom Flavor	8.12 – 9.0
8	Like very much Mushroom Flavor	7.23 – 8.11
7	Like moderately Mushroom Flavor	6.34 – 7.22
6	Like slightly Mushroom Flavor	5.45 – 6.33
5	Neither like nor dislike Mushroom Flavor	4.56 – 5.44
4	Dislike slightly Mushroom flavor	3.67 – 4.55
3	Dislike moderately Mushroom flavor	2.78 – 3.66
2	Dislike very much Mushroom flavor	1.89 – 2.77
1	Dislike extremely Mushroom flavor	1.00 – 1.88

Table 1.3. *Nine-Point Hedonic Soring Scale in terms of Texture.*

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
9	Like Extremely firm Texture	8.12 – 9.0
8	Like very much firm Texture	7.23 – 8.11
7	Like moderately firm Texture	6.34 – 7.22
6	Like slightly firm Texture	5.45 – 6.33
5	Neither like nor dislike firm Texture	4.56 – 5.44
4	Dislike slightly firm Texture	3.67 – 4.55
3	Dislike moderately firm Texture	2.78 – 3.66
2	Dislike very much firm Texture	1.89 – 2.77
1	Dislike extremely firm Texture	1.00 – 1.88

Table 1.4. *Nine-Point Hedonic Soring Scale in terms of the General Acceptability of the consumers.*

Scoring	Verbal interpretation/Response Categories	Score Interval
9	Like Extremely	8.12 – 9.0
8	Like Very Much	7.23 – 8.11
7	Like Moderately	6.34 – 7.22
6	Like Slightly	5.45 – 6.33
5	Neither Like nor Dislike	4.56 – 5.44
4	Dislike Slightly	3.67 – 4.55
3	Dislike Moderately	2.78 – 3.66
2	Dislike Very Much	1.89 – 2.77
1	Dislike Extremely	1.00 – 1.88

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RESULTS AND DISCUSSION

In terms of the Colors of the Oyster Mushroom Patty, as shown in Table 2.0, Treatment A (300g of Oyster Mushroom) got the highest mean of 7.29 with a score of 8 based on the nine-point hedonic scoring scale, which is equivalent to the verbal interpretation of Dark Brown color. In comparison, Treatment B (350g of Oyster Mushroom) with a mean of 6.68, and Treatment C (400g of Oyster Mushroom) with a mean of 6.70 have the same score of 7 based on the nine-point hedonic scoring scale which is equivalent to the verbal interpretation of Medium Brown.

Table 2.0. Sensory characteristics of different treatments with replication in terms of Color.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	7.47	7.9	6.5	21.87	7.29	Dark Brown
B	6.47	7.3	6.27	20.04	6.68	Medium Brown
C	6.43	7.7	5.9	20.1	6.70	Medium Brown

In terms of the Aroma of Oyster Mushroom Patty, as shown in Table 2.1, all treatments got the same score of 6 based on the nine-point hedonic scoring scale, equivalent to the verbal interpretation of like Slightly Mushroom Aroma.

Table 2.1. Sensory characteristics of different treatments in terms of Aroma.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	5.53	6.57	5.7	17.8	5.93	Like Slightly Mushroom Aroma
B	5.37	7.07	5.9	18.34	6.11	Like Slightly Mushroom Aroma
C	5.43	7.33	5.47	18.23	6.08	Like Slightly Mushroom Aroma

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In terms of the Taste of Oyster Mushroom Patty, as shown in Table 2.2, Treatment B (350g of Oyster Mushroom) with a mean of 6.44 and Treatment C (400g of Oyster Mushroom) with a mean of 6.53 got the highest score of 7 based on the nine-point hedonic scoring scale which is equivalent to verbal interpretation of Like Moderate Mushroom Flavor and Treatment A (300g of Oyster Mushroom) with mean of 6.18 got the lowest score of 6 based on the nine-point hedonic scoring scale which is equivalent to verbal interpretation of like Slightly Mushroom Flavor.

Table 2.2. Sensory characteristics of different treatments in terms of Taste.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	5.47	6.97	6.1	18.54	6.18	Like Slightly Mushroom Flavor
B	5.37	8.23	5.73	19.33	6.44	Like Moderate Mushroom Flavor
C	5.4	8.23	5.97	19.6	6.53	Like Moderate Mushroom Flavor

In terms of the texture of the Oyster Mushroom Patty, as shown in Table 2.3, Treatment B (350g of Oyster Mushroom) with a mean of 6.44, and Treatment C (400g Oyster Mushroom) with a mean of 6.65 got the highest score of 7 based on the nine-point hedonic scoring scale which is equivalent to verbal interpretation of like moderately firm texture. In comparison, Treatment A (300g of Oyster Mushroom) with a mean of 5.89, got the lowest score of 6 based on the nine-point hedonic scoring scale, equivalent to verbal interpretation of like slightly firm texture.

Table 2.3. Sensory characteristics of different treatments in terms of Texture.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	5.67	6.57	5.43	17.67	5.89	Like Slightly Firm Texture
B	5.37	8.23	5.73	19.33	6.44	Like Moderately Firm Texture
C	6.17	7.77	6	19.94	6.65	Like Moderately Firm Texture

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For the General Acceptability of Oyster Mushroom Patty, as shown in Table 2.4, Treatment C (400g of Oyster Mushroom) got the highest mean of 6.52 with a score of 7 based on the nine-point hedonic scoring scale, which is equivalent to the verbal interpretation of Like Moderate. In comparison, Treatment A (300g of Oyster Mushroom) with a mean of 6.13, and Treatment B (350g of Oyster Mushroom) with a mean of 6.27, have the same score of 6 based on the nine-point hedonic scoring scale, which is equivalent to the verbal interpretation of which is equivalent to Like Slightly.

Table 2.4. Sensory characteristics of different treatments in terms of General Acceptability.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	5.37	7.3	5.73	18.4	6.13	Like Slightly
B	5.4	7.87	5.53	18.8	6.27	Like Slightly
C	6	7.63	5.93	19.56	6.52	Like moderately

SENSORY ANALYSIS OF VARIANCE (ANOVA) RESULT

Descriptive statistics of the sensory characteristics in terms of color, aroma, taste, texture, and general acceptability using a nine-point hedonic scale.

In Table 3.0. Using statistical analysis and one-way ANOVA, reveals the difference in the color of the mushroom patty. Results disclosed that there is a significant difference existed in the color of the mushroom patty.

Table 3.0. One-way ANOVA results of the sensory characteristics in terms of Color.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	93.8	2	46.9	46.57	4.46E-18
With groups:	268.867	267	1.00699	Permutation p (n=99999)	
Total:	362.667	269	1.00E-05		

*The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.

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In Table 3.1. Using statistical analysis and one-way ANOVA, reveals the difference in the aroma of the mushroom patty. Results disclosed that there is a significant difference existed in the aroma of the mushroom patty.

Table 3.1 One-way ANOVA results of the sensory characteristics in terms of Aroma.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	124.052	2	62.0259	45.94	7.14E-18
With groups:	360.5	267	1.35019	Permutation p (n=99999)	
Total:	484.552	269	1.00E-05		

*The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.

In Table 3.2. Using statistical analysis and one-way ANOVA, reveals the difference in the taste of the mushroom patty. Result disclosed that there is a significant difference existed in the taste of the mushroom patty.

Table 3.2 One-way ANOVA results of the sensory characteristics in terms of Taste.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	286.763	2	143.381	86.38	1.17E-29
With groups:	443.178	267	1.65984	Permutation p (n=99999)	
Total:	729.941	269	1.00E-05		

*The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.

In Table 3.3. Using statistical analysis and one-way ANOVA, reveals the difference in the texture of the mushroom patty. Result disclosed that there is a significant difference existed in the texture of the mushroom patty.

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Table 3.3 One-way ANOVA results of the sensory characteristics in terms of Texture.

	Sum Squares	df	Mean Square	F	Sig.
Between groups:	65.4889	2	32.7444	19.76	9.90E-09
With groups:	442.378	267	1.65685	Permutation p (n=99999)	
Total:	507.867	269	1.00E-05		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.4. Using statistical analysis and one-way ANOVA, reveals the difference in general acceptability of the mushroom patty. Result disclosed that there is a significant difference existed in general acceptability of the mushroom patty.

Table 3.4 One-way ANOVA results of the sensory characteristics in terms of General Acceptability.

	Sum Squares	df	Mean Square	F	Sig.
Between groups:	226.496	2	113.248	73.57	3.55E-26
With groups:	410.989	267	1.53928	Permutation p (n=99999)	
Total:	637.485	269	1.00E-05		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

CONSUMER ANALYSIS (ANOVA) RESULT

As presented in Table 4.0, the result of the sensory characteristic of different Respondent in terms of General Acceptability, the Oyster Mushroom Patty are most acceptable in employee respondents and have a higher mean of 8.44 and least acceptable with students in elementary level respondent with a mean of 6.8.

Table 4.0. The mean consumer acceptability using nine-point hedonic scale.

Respondents	Total	Mean	Verbal Interpretation
Elementary students	170	6.8	Like Moderately
High School students	196	7.84	Like Very Much
College students	180	7.2	Like Moderately
Employee	211	8.44	Like Very Much

**Legend: 1= Dislike Extremely, 2= Dislike Very Much, 3= Dislike Moderately, 4= Dislike Slightly, 5= Neither Like nor Dislike, 6= Like Slightly, 7= Like Moderately, 8= Like Very Much, 9= Like Extremely*

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Using statistical analysis and one-way ANOVA, results showed in Table 4.1 that there is a significant difference in consumer acceptability between four group respondents.

Table 4.1. *The statistics of consumer acceptability using nine-point hedonic scale*

	Sum Squares	of df	Mean Square	F	Sig.
Between groups:	38.99	3	12.9967	15.31	3.25E-08
With groups:	81.52	96	0.849167	Permutation p (n=99999)	
Total:	120.51	99	1.00E-05		

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

CONCLUSION

Since the sensory qualities of the different levels of oyster mushrooms in making patty have a pleasing, authentic, and delicious quality, it is therefore concluded that the oyster mushroom can be used in making patty. All levels of oyster mushroom can be utilized in making patty and has the potential for product development and commercialization. The cost of production of the mushroom patty is cheaper than the cost of a commercial available patty.

The product has high consumer preference and is moderately acceptable. As to consumer's sensory evaluation, the products have a significant difference in the color, aroma, taste, texture, and general acceptability of oyster mushroom patty.

RECOMMENDATIONS

The three formulations of oyster mushrooms used in the study are recommended in making mushroom patty, for they have the potential for product development and commercialization.

The oyster mushroom patty's aroma, color, taste, and texture can still be enhanced by considering other modification processes or adding spices and herbs.

For further study, it can be recommended that other researchers come up with a new product using plant-based ingredients.

For consumers, this research recommends consuming healthy patty for healthy living.

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Output (Indicate the specific products, processes, or services that the project has produced; quantify when possible.)

Mushroom Patty

Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes, or services transferred as extension activities)

Mushroom Patty Product

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